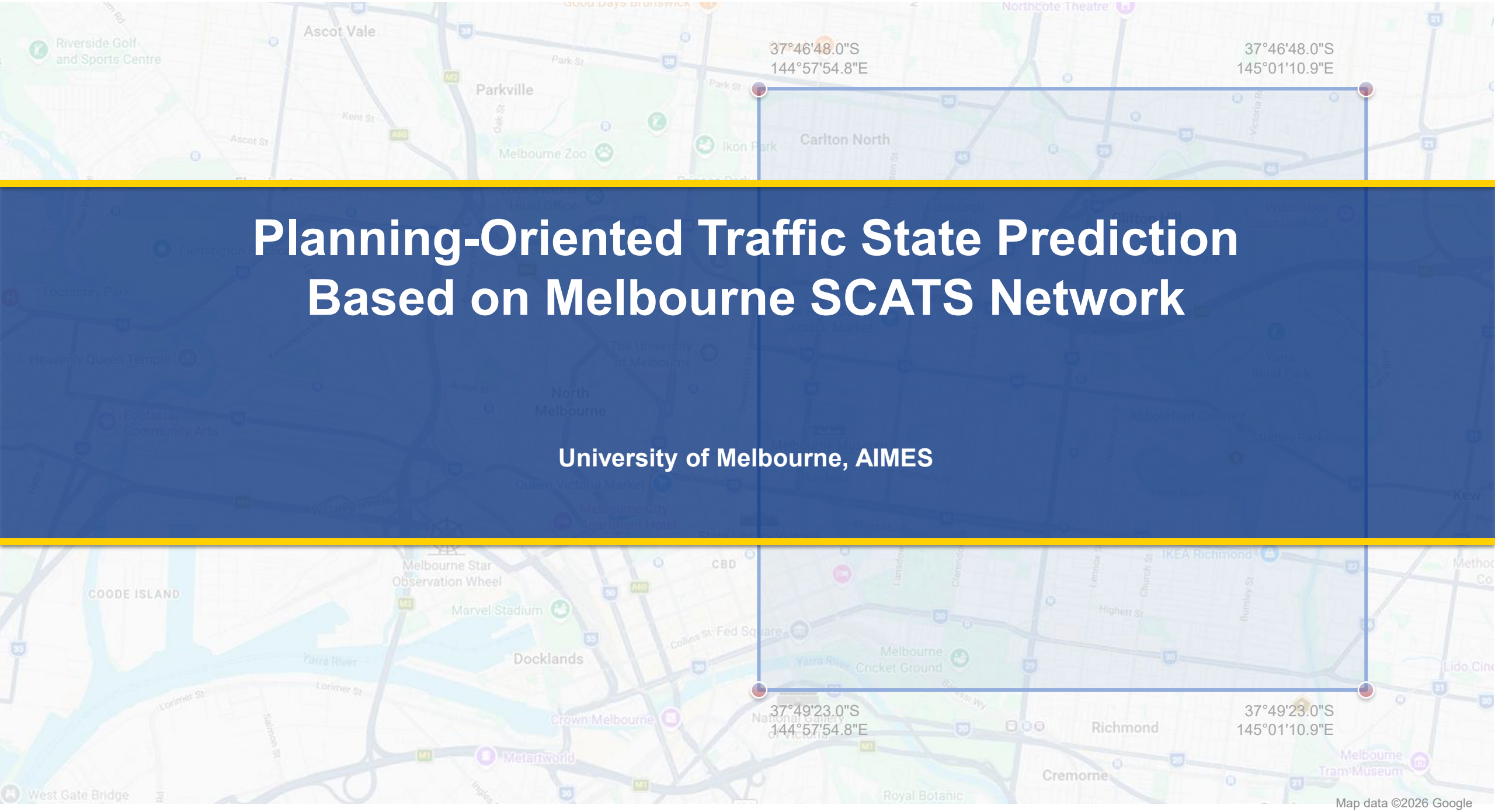



# Planning-Oriented Traffic State Prediction Based on Melbourne SCATS Network

**University of Melbourne, AIMES**

# Background and Study Area

Melbourne SCATS signal network — Victoria Open Data Portal

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## Research objective

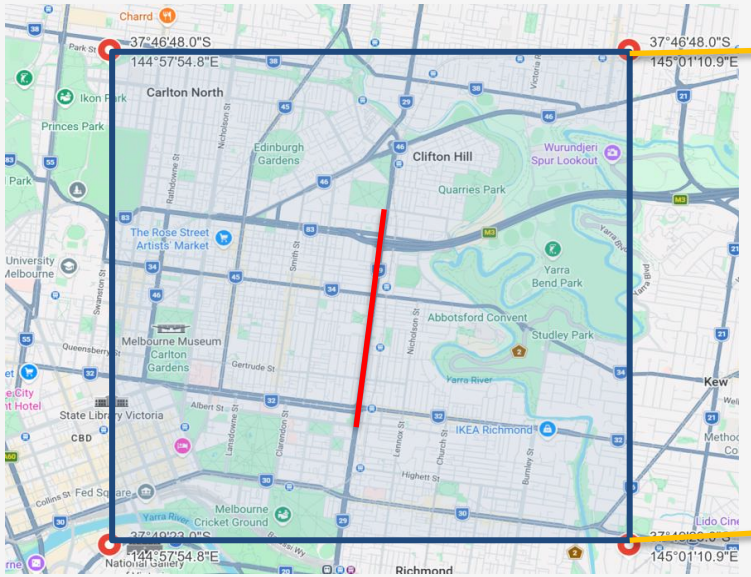
Traffic demand is shaped not only by history, but also by the surrounding built environment visible in satellite imagery.

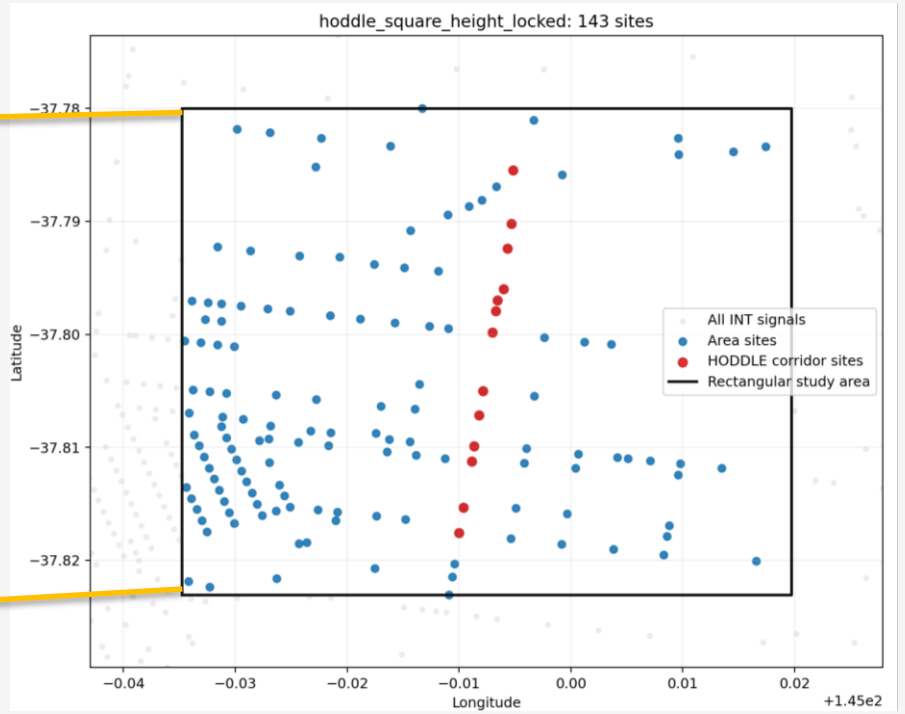
This enables a planning use case: when a site or parcel changes, the updated land-use pattern can be translated into a future traffic-flow estimate.

### Current study area

A rectangular study area centered on the Hoddle corridor. (Near Unimelb)
143 SCATS sites; the model uses the 81 with CLUE land-use data.

- Traffic Signal Volume Data — Victorian Open Data Portal
- Victoria Traffic Signals — site coordinates & metadata
- Time range: 2014-2024, 15-min slots (2020 and 2021 left out)
- Satellite imagery: Nearmap, one survey a year



# Proposed Multimodal Framework

Route 1 learns visual semantics from satellite images; Route 2 extends with network and traffic topology JSON embeddings.

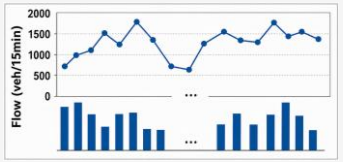
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## Inputs

**1 Satellite imagery**



**2 15-min SCATS traffic flow data**




- site
- date
- time slot (15-min)
- historical flow

**3 Land use information**

*Site-year alignment / temporal indexing / log-ratio target construction*

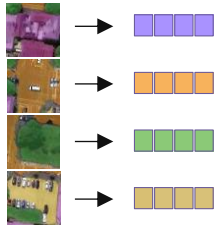
## Two-Route Multimodal Modelling

### Route 1: Visual Semantic Route

**SAM 3.1 semantic region proposal**



**Per-region DINO embedding**



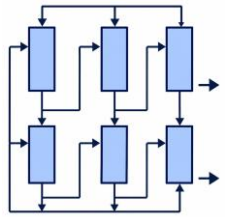
**Aggregation by class**

1, 2, 3, ⋮, k

**Site-year visual feature table**

| site_id | year | … | $f_1$ | … | f_d |
|---|---|---|---|---|---|
| 1001 | 2021 | … | … | … | … |
| 1001 | 2022 | … | … | … | … |
| ⋮ | ⋮ | ⋱ | ⋮ | ⋮ | ⋮ |
| 2050 | 2024 | … | … | … | … |

**Day-TCN predictor**



*Target: log ratio to the 2022 historical profile*

**24-d dynamic temporal context**

calendar cyclical · weekend / year · holiday / event · historical base profile

*Route 1 validates and improves the visual frontend; Route 2 adds LLM-derived topology context.*

*shared visual frontend*

### Route 2: LLM-Enhanced Multimodal Route

**A Visual embedding** — SAM-DINO visual features …

**B Network topology JSON** — { } JSON

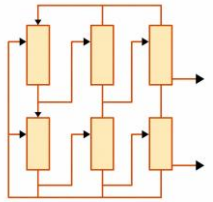
- site_id
- degree
- Neighbors
- edge_weights
- 2-hop summary

*LLM embedding: Qwen3-Embedding hidden layer pooling + PCA compression*

LLM embedding model → Net JSON embedding …

**24-d dynamic temporal context**

**Fusion Day-TCN / multimodal predictor**



## Outputs

**Predicted 15-min traffic flow**



**Log ratio → volume**



**Evaluation:** MAE, RMSE, MAPE, GEH, GEH<5%

# Route 1 and Route 2

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**Inputs**
Nearmap satellite image
500 m x 500 m per site
CLUE land use
2022 historical profile
Calendar and holidays

**Route1**
SAM 3: 4 land-cover classes
DINOv3 features, PCA to 16-d
Class aggregation: 135-d
Output: visual feature

**Route2**
Route1 visual feature
+ network JSON embedding
(Qwen3 embedding, PCA 64)
Output: site-year feature

**Predictor**
Temporal convolution (TCN)
Imagery, network, land use
24-d calendar & holidays
Ratio to 2022 profile
Output: 96 x 15-min flows

Spring St / Flinders Ln, 2014
CBD: satellite image



Spring St / Flinders Ln, 2014
SAM 3 on the whole 500 m image



Alexandra Pde / Wellington St, 2020
Collingwood: satellite image



Alexandra Pde / Wellington St, 2020
SAM 3 on 150 m tiles



Road  Parking / paved  Roof  Vegetation

# Experimental Results

Ablation experiments: test year 2024, 81 intersections, mean of 5 random seeds.

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| Model | MAE | RMSE | GEH Median | GEH<5 (%) | GEH<10 (%) |
|---|---|---|---|---|---|
| Time only | 62.22 | 127.7 | 4.14 | 57.2 | 81.6 |
| Route 1: + imagery | 60.55 | 122.5 | 3.96 | 58.8 | 82.0 |
| **Route 2: + imagery + network** | **60.14** | **122.3** | **3.93** | **59.1** | **82.3** |

Satellite imagery lowers MAE by 1.67 ($p = 0.002$), in all 5 random seeds.
Route 2 (imagery + road network) is the best on all five metrics, with MAE 3.3% below time only.

# Land-Use Data: CLUE

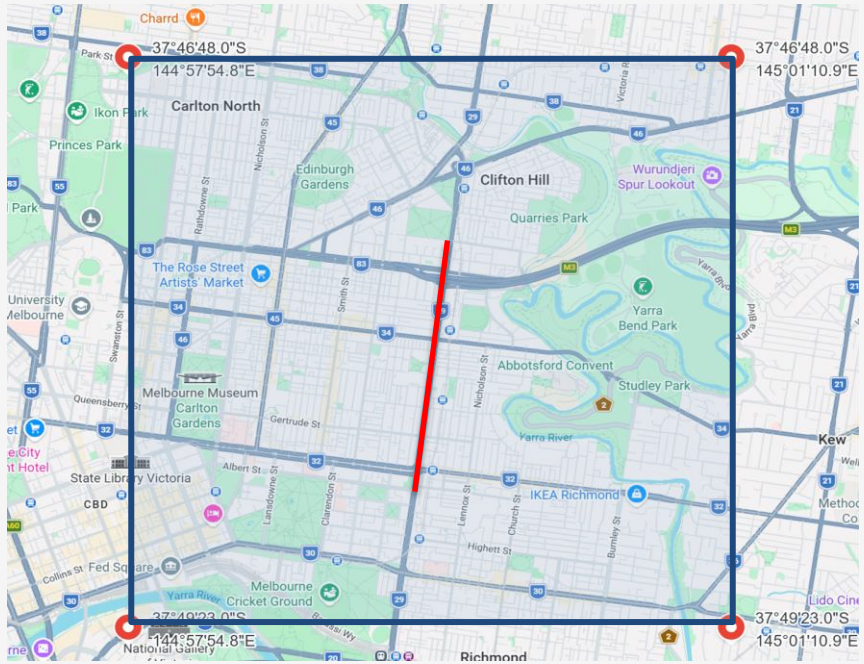
Verify the effectiveness of adding building features. only-time, route1 and route2

## Add building feature in traffic prediction

1. Dataset: City of Melbourne: CLUE (Census of Land Use and Employment)
2. Building height data only in City of Melbourne
3. Year coverage: 2002—2024
4. Problem of the satellite image from City of Melbourne: Too much duplicate data, before 2020. (Change another satellite image dateset: Nearmap)

train 2014-2019 + 2022, val 2023, test 2024

Previous work—traffic prediction




2020-2026

2020-2024(5 years, 3 years training)

# Land-Use (Building) Features

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## Add building feature in traffic prediction

### 43-dim aggregate statistics

Everything inside the site area is summarised into one number per quantity

| 1 | Quantity | Buildings |
|---|---|---|
| 9 | Height | floors_mean / max / sum / p90, tall_over_10, tall_over_30, floors_sum_inner, floors_max_inner, floors_sum_weighted |
| 4 | Distance | buildings_inner, buildings_weighted, nearest_over_10_m, nearest_over_30_m |
| 26 | Land use | share_* × 13, floorshare_* × 13 |
| 3 | Other | bicycle_spaces, built_last_5y, median_age |

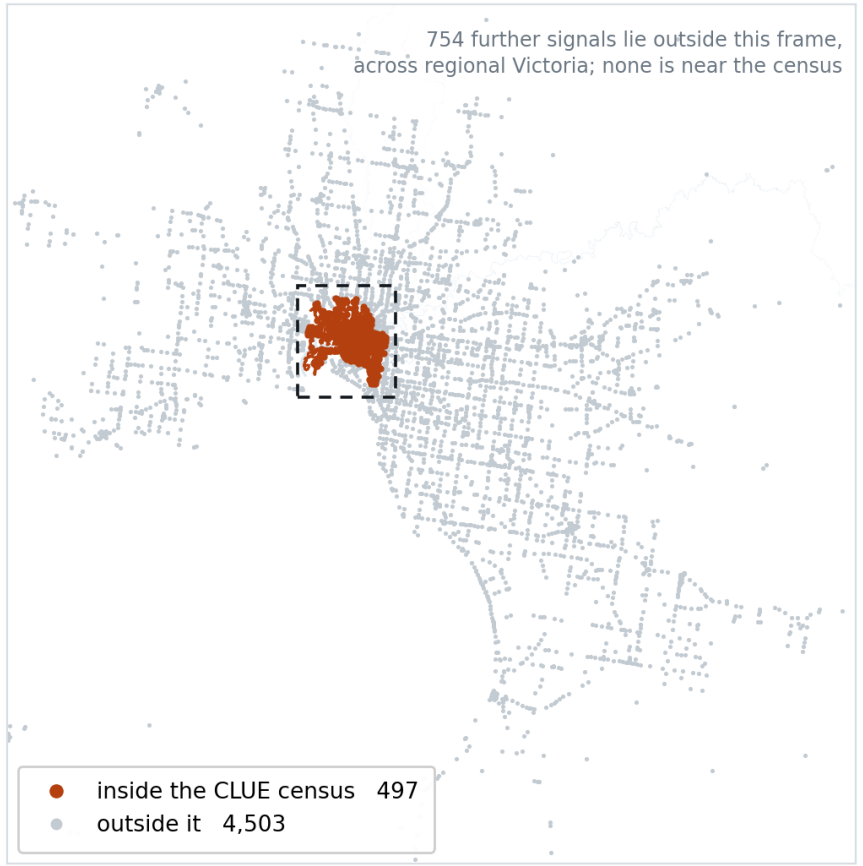
# Validation in Different Regions

Validate the effectiveness of the model in different regions.

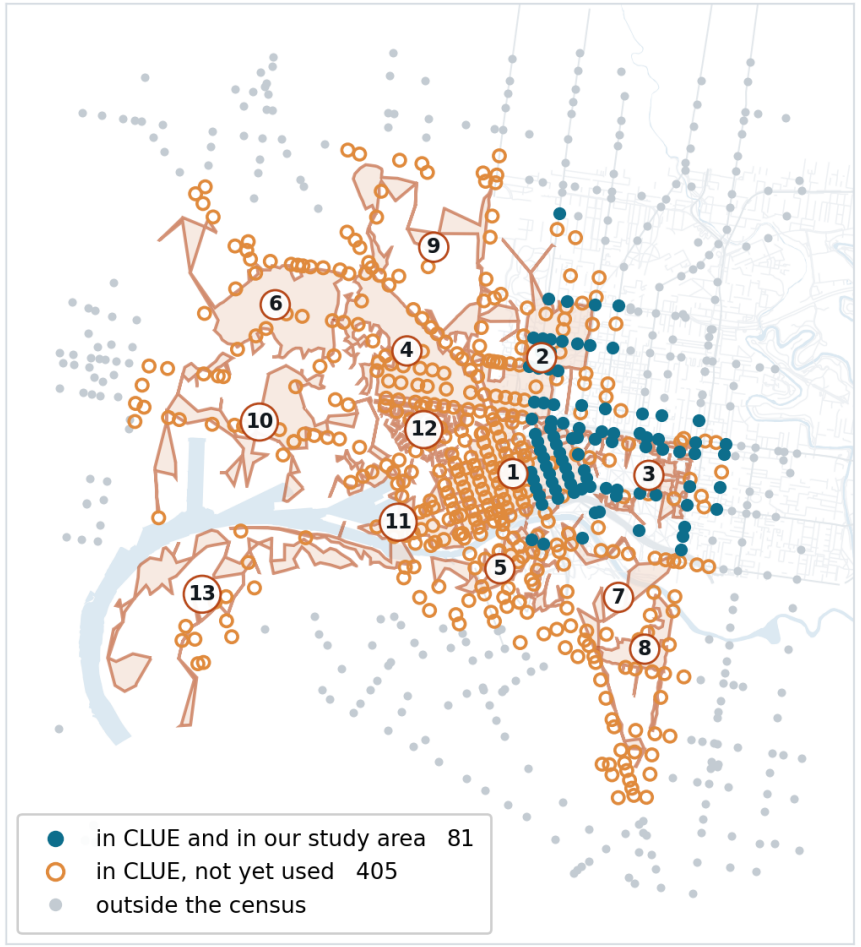
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## The land-use census reaches one tenth of the signal network

Greater Melbourne — 4,246 of the state's 5,000 SCATS signals




CLUE ∩ SCATS — the 13 small areas and the signals in them




| CLUE small area | signals |
|---|---|
| 1 Melbourne (CBD) | 116 |
| 2 Carlton | 60 |
| 3 East Melbourne | 48 |
| 4 North Melbourne | 48 |
| 5 Southbank | 45 |
| 6 Kensington | 30 |
| 7 Melbourne (Remainder) | 26 |
| 8 South Yarra | 26 |
| 9 Parkville | 24 |
| 10 W Melbourne (Industrial) | 24 |
| 11 Docklands | 22 |
| 12 W Melbourne (Residential) | 16 |
| 13 Port Melbourne | 12 |
| **total** | **497** |

# Results: Land Use and a Wider Region

Test year 2024, mean of 5 random seeds (final report, Tables 4.12 and 4.14).

| | MAE | GEH_median |
|---|---|---|
| Time only | 62.22 | 4.14 |
| Time + land use | 61.36 | **3.94** |
| | | |
| Route 1 | 60.55 | 3.96 |
| Route 1 + land use | 59.51 | **3.88** |
| | | |
| Route 1 (371 sites) | 59.11 | 4.13 |
| Route 1 + land use (371 sites) | 59.25 | **4.11** |
| | | |
| Route 2 | 60.14 | 3.93 |
| Route 2 + land use | 59.18 | **3.84** |
| | | |
| Route 2 (371 sites) | 59.24 | 4.13 |
| Route 2 + land use (371 sites) | 58.76 | **4.11** |

# Land-Use Sensitivity Analysis

Sensitivity analysis of building attribute features.

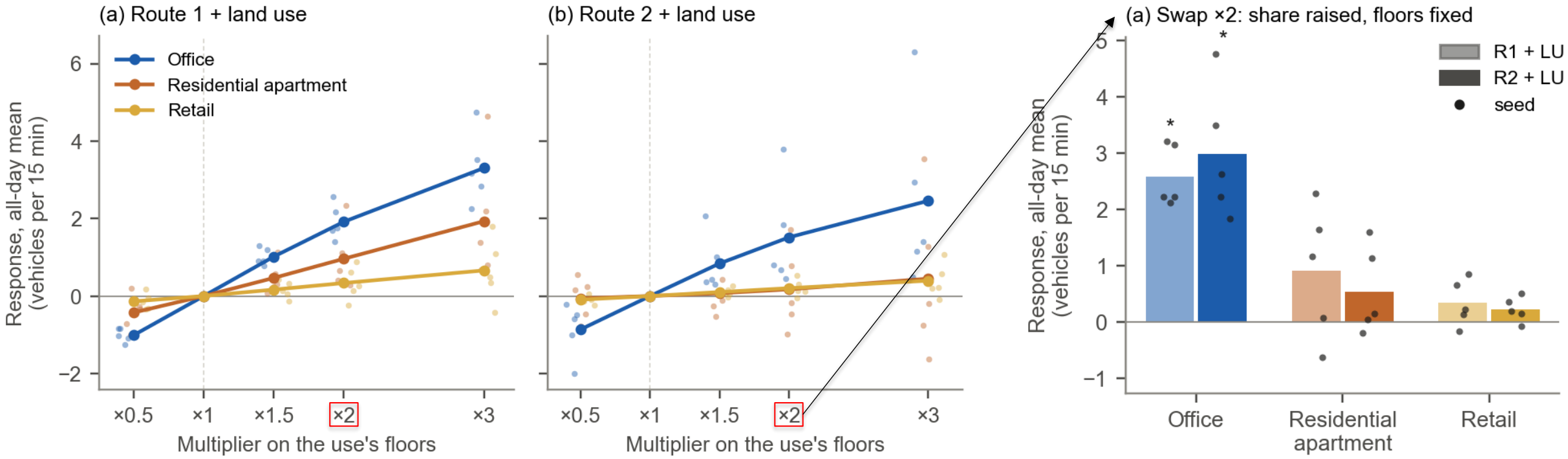
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New building

For example: office space from 10->20
Then total space from 50->60

Change building attribute from other 12 types buildings

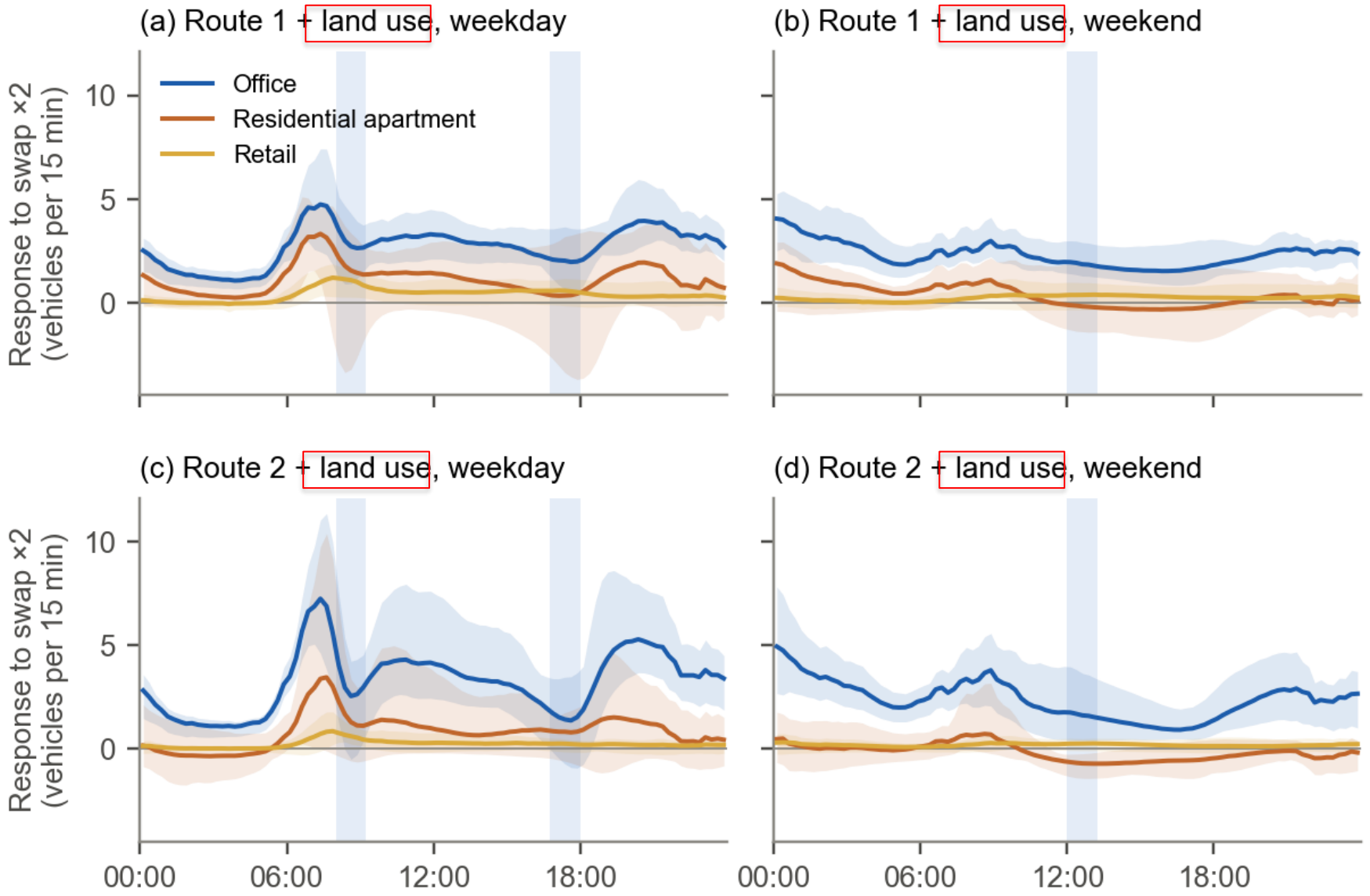
For example: office space from 10->20
Then total space remains 50



# Land-Use Sensitivity: Time of Day

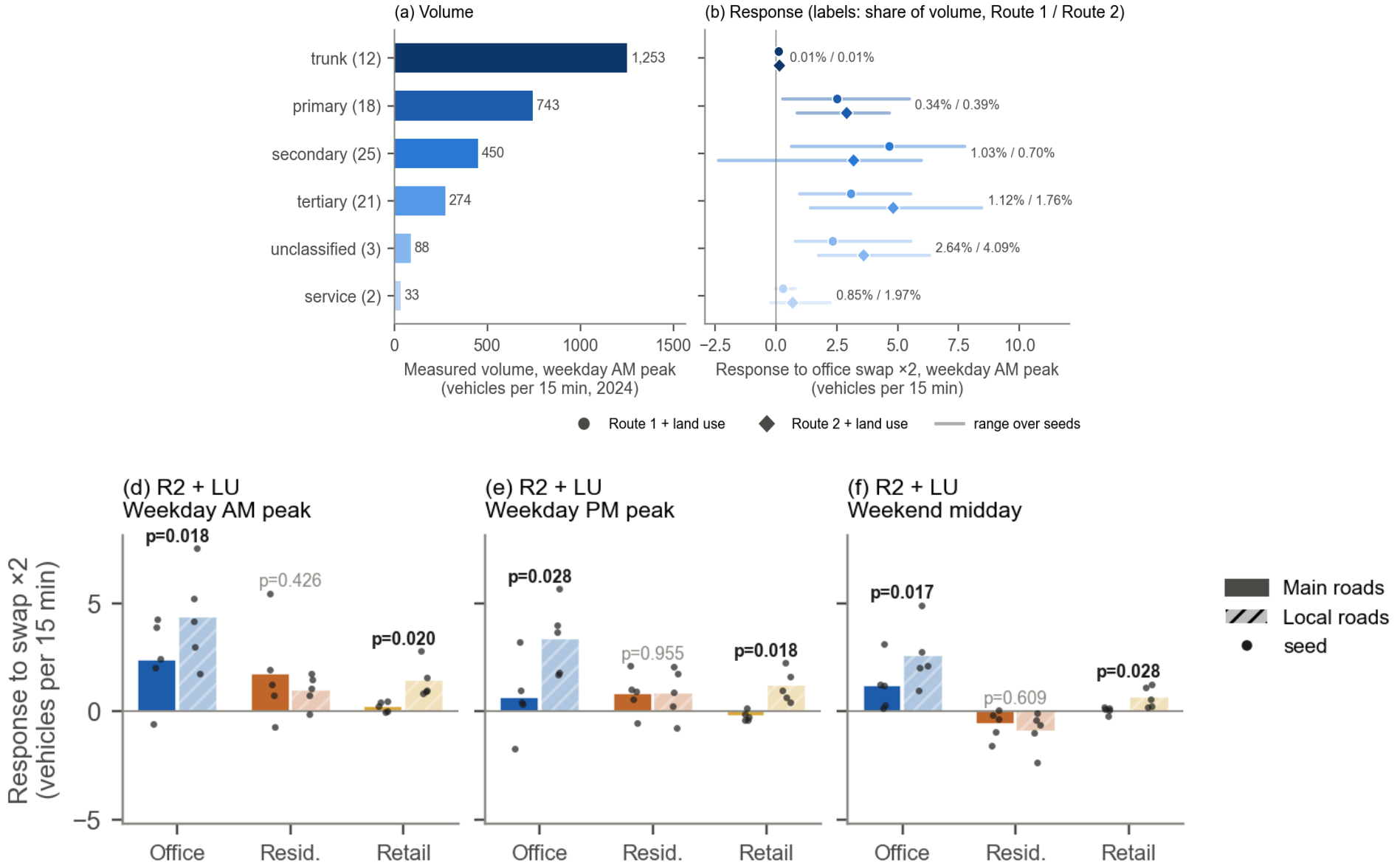
Sensitivity analysis of building attribute features.

Swap
Floorspace * 2



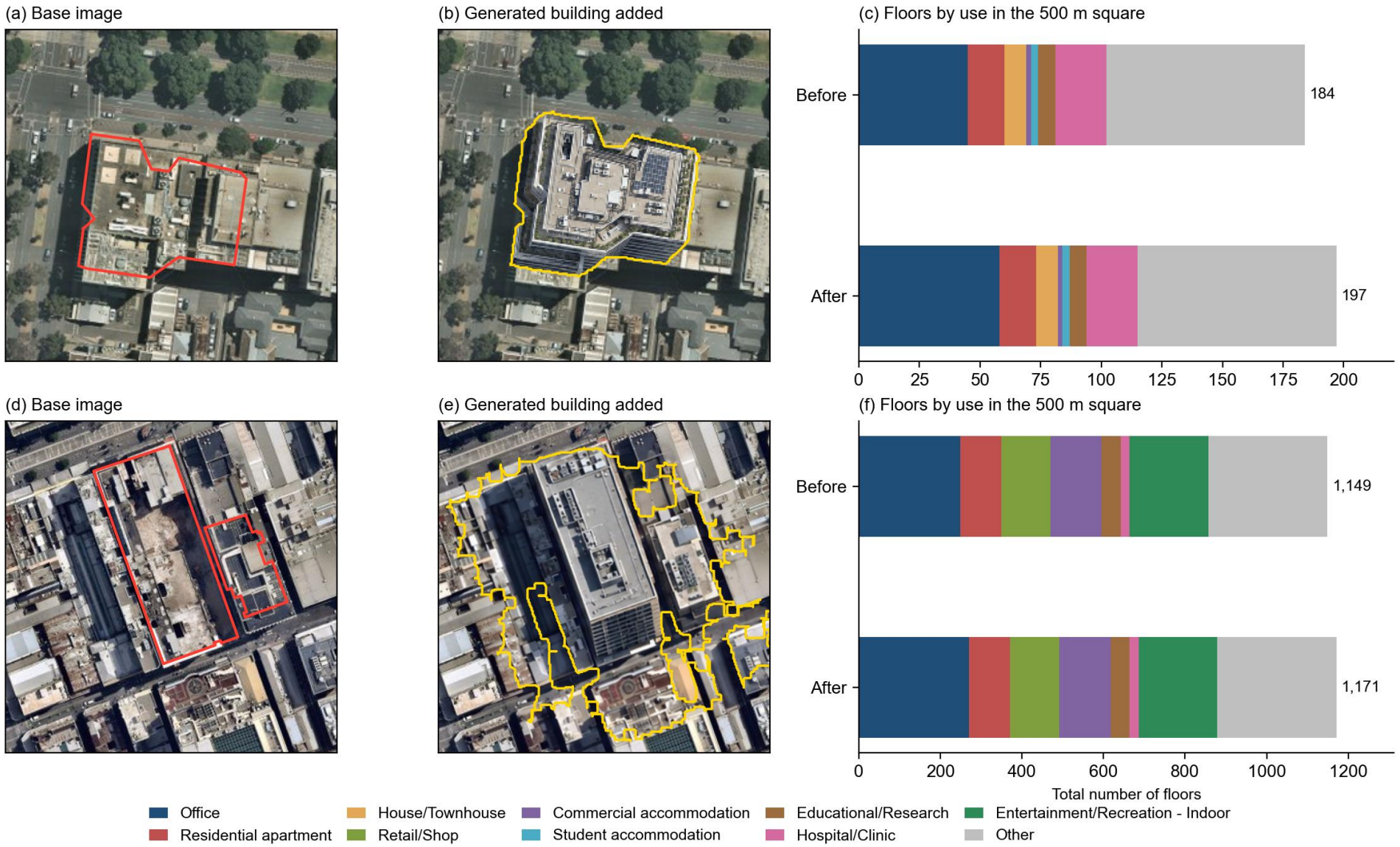
# Land-Use Sensitivity: Road Class

Sensitivity analysis of “all roads” and “only main roads”



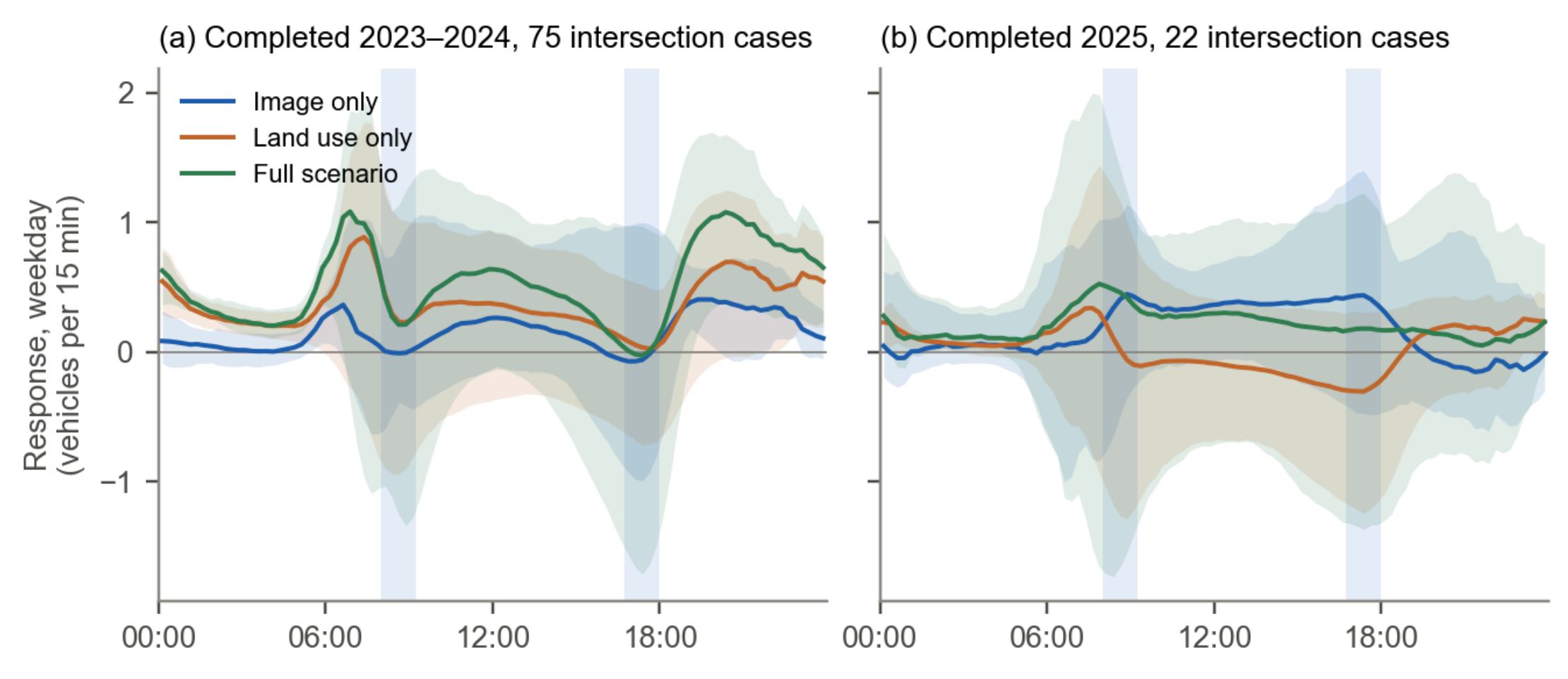
# Generated Imagery: Test Cases

Real developments completed in 2023-2025, generated on imagery from before construction.



# Generated Imagery in Traffic Prediction

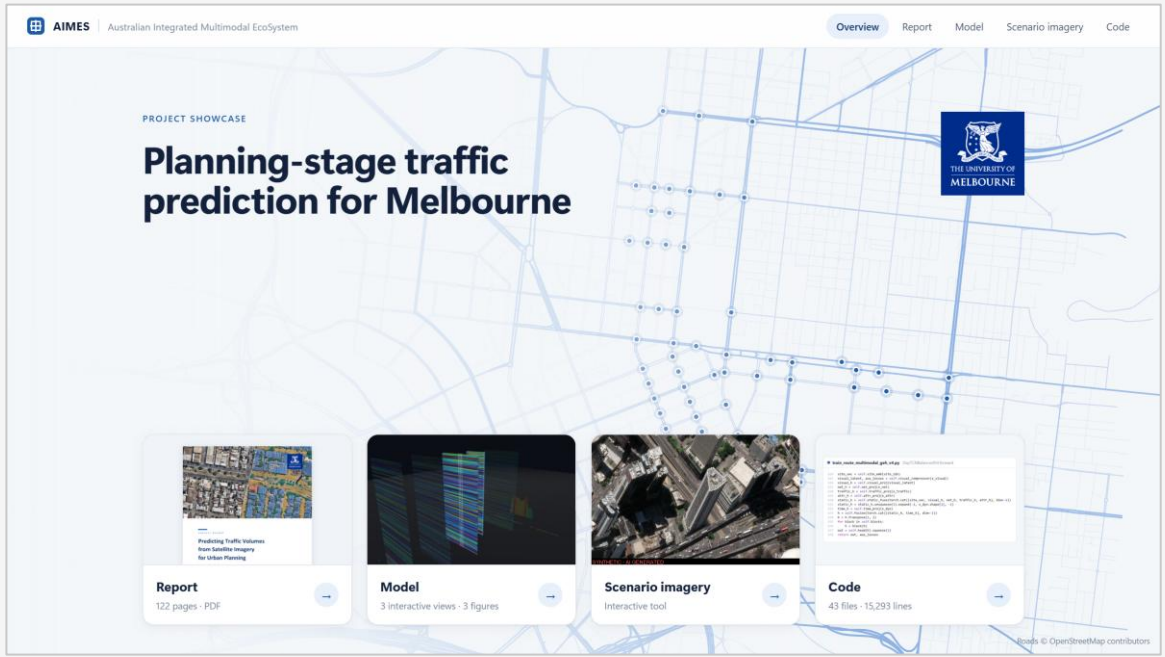
Response of Route 2 + land use to each case against no development (weekdays, 5 seeds).

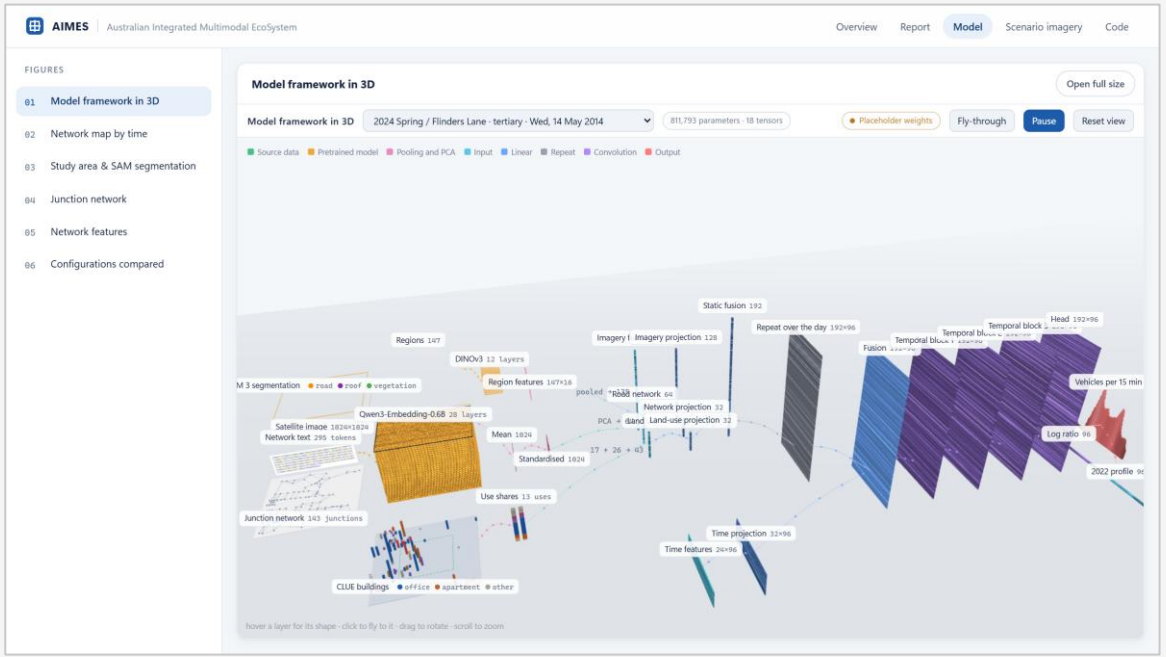



Accuracy with generated imagery: median GEH 3.50-4.06, MAE changes by at most 0.15 veh/15 min.

Full scenario (generated building and updated land use): +0.49 and +0.19 veh/15 min.

# Project Website




Project website: [cjsrc.github.io/Melbourne_traffic_showcase](cjsrc.github.io/Melbourne_traffic_showcase)

Scenario imagery tool: [cjsrc.github.io/Melbourne_traffic_showcase/tool.html](cjsrc.github.io/Melbourne_traffic_showcase/tool.html)

Report (122-page PDF), interactive model views, scenario imagery examples and the project code.